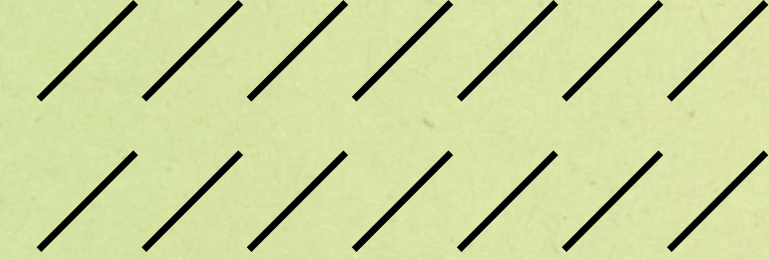


SUPPLY CHAIN & PROCUREMENT ANALYTICS PROJECT

RISK-CONSTRAINED PROCUREMENT OPTIMIZATION USING SUPPLIER RELIABILITY ANALYTICS

By - Neelaksh Lohani

BUSINESS UNDERSTANDING



Stakeholder:

Procurement Operations Manager responsible for overseeing supplier reliability, delivery performance, and sourcing cost efficiency across procurement regions.

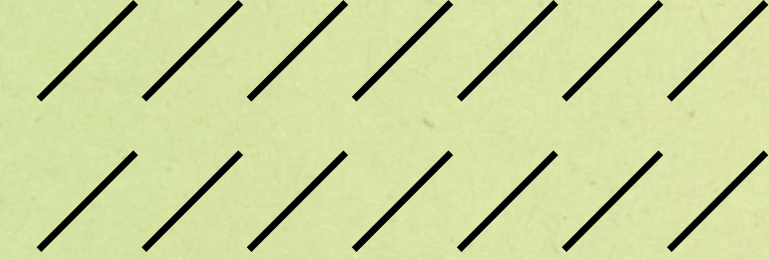
Business Context:

The organization procures products from multiple sourcing regions using different shipment modes, each incurring varying delivery lead time and procurement cost.

Operational inefficiencies in supplier selection and logistics strategy can result in:

- Increased procurement cost per unit
- Delayed deliveries
- SLA violations
- Supplier reliability issues
- Reduced sourcing efficiency

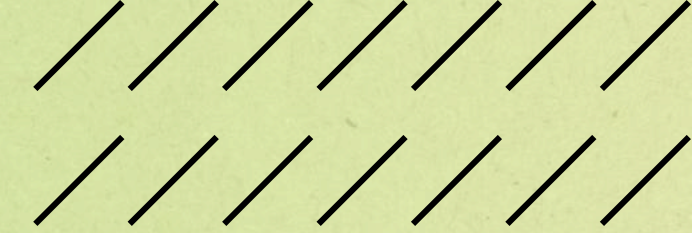
BUSINESS UNDERSTANDING



Key Business Decisions Required:

- Which sourcing regions exhibit higher supplier risk exposure?
- Are delivery delays impacting SLA compliance?
- Which shipment modes are cost-inefficient under varying risk conditions?
- Should procurement strategy vary across product categories?
- How can sourcing reliability be improved without increasing procurement cost?

KEY METRICS USED



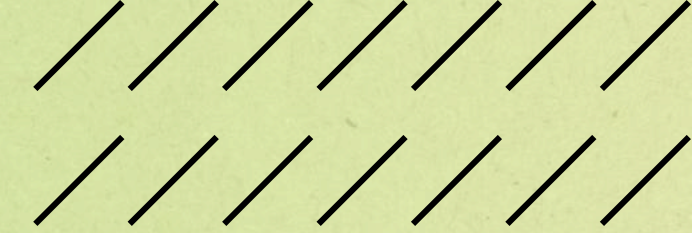
Fulfillment Performance Metrics:

- Supplier Reliability Metrics:
- Supplier Reliability Risk Score
- Delivery Delay Risk (%)
- SLA Violation Rate (%)
- Lead Time Stability Index

Procurement Cost Metrics:

- Risk Adjusted Procurement Cost
- Region-wise Procurement Cost
- Shipment Mode-wise Cost

DATASET OVERVIEW

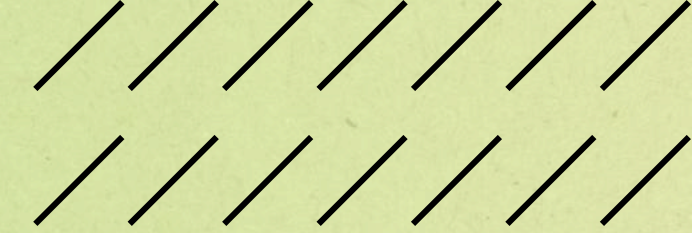


The analysis utilizes the DataCo Smart Supply Chain Dataset, capturing order-level procurement and logistics operations across sourcing regions.

Dataset Contains:

- Order Date
- Shipment Date
- Shipment Mode
- Product Category
- Procurement Region
- Delivery Status
- Shipment Lead Time
- Procurement Cost
- Profit per Order

DATASET OVERVIEW



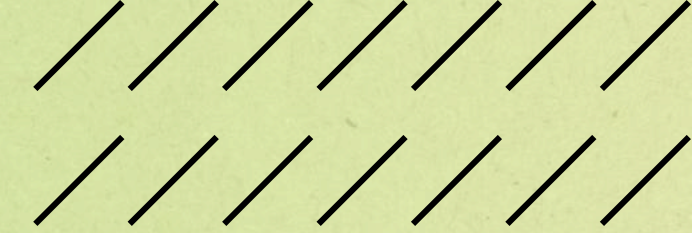
Each record represents:

A single procurement transaction containing product, regional sourcing, and shipment information.

This dataset enables analysis of:

- Supplier delivery performance
- Shipment mode reliability
- Region-wise sourcing risk
- Procurement cost distribution
- SLA compliance

BUSINESS PROBLEMS



Operational Challenges:

- Supplier delivery delays across sourcing regions
- Shipment mode-dependent fulfillment reliability
- Lead time variability impacting procurement consistency

Financial Challenges:

- Rising procurement cost per unit
- Cost-risk tradeoff across shipment modes
- Risk-prone sourcing strategies impacting operational efficiency

Analytical Objective:

To identify procurement risk exposure and recommend sourcing strategies that improve delivery reliability while minimizing procurement cost.

DATA CLEANING & TRANSFORMATION (PostgreSQL)

Transformation Performed in PostgreSQL:

The transactional procurement dataset was cleaned and transformed to standardize lead time metrics and enable reliable risk computation.

1.) Handling Mixed Date Formats

```
CREATE TABLE staging_layer.procurement_clean AS
SELECT
*,
CAST(order_date AS DATE) AS order_date_clean,
CAST(shipping_date AS DATE) AS shipping_date_clean
FROM raw_layer.dataco_supplychain_raw;
```

2.) COMPUTING ACTUAL LEAD TIME

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN actual_lead_time INT;

UPDATE staging_layer.procurement_clean
SET actual_lead_time =
shipping_date_clean - order_date_clean;
```


DATA CLEANING & TRANSFORMATION (PostgreSQL)

3.) LEAD TIME VARIANCE CALCULATION

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN lead_time_variance INT;

UPDATE staging_layer.procurement_clean
SET lead_time_variance =
actual_lead_time - days_for_shipment_scheduled;
```

4.) DELIVERY DELAY FLAG CREATION

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN delay_flag INT;

UPDATE staging_layer.procurement_clean
SET delay_flag =
CASE
WHEN lead_time_variance > 0 THEN 1
ELSE 0
END;
```


DATA CLEANING & TRANSFORMATION (PostgreSQL)

5.) SLA FAILURE CLASSIFICATION

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN sla_failure_flag INT;

UPDATE staging_layer.procurement_clean
SET sla_failure_flag =
CASE
WHEN delivery_status = 'Late delivery' THEN 1
ELSE 0
END;
```

6.) COST PER UNIT CALCULATION

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN cost_per_unit NUMERIC;

UPDATE staging_layer.procurement_clean
SET cost_per_unit =
order_item_total / NULLIF(order_item_quantity,0);
```

7.) PROCUREMENT PROFIT MARGIN

```
ALTER TABLE staging_layer.procurement_clean
ADD COLUMN profit_margin NUMERIC;

UPDATE staging_layer.procurement_clean
SET profit_margin =
order_profit_per_order / NULLIF(order_item_total,0);
```

8.) REMOVAL OF LEAD TIME OUTLIERS

```
DELETE FROM staging_layer.procurement_clean
WHERE actual_lead_time < 0
OR actual_lead_time > 30;
```


KPI ENGINEERING (Power BI – DAX)

Operational KPIs were created using DAX measures to evaluate supplier reliability and procurement risk exposure across sourcing regions and shipment modes.

1.) Supplier Risk Index

```
1 Supplier Risk Index =  
2 0.35 * [Delay Probability]  
3 + 0.25 * [SLA Failure Rate]  
4 + 0.20 * AVERAGE(fact_procurement_orders[region_delay_probability])  
5 + 0.20 * AVERAGE(fact_procurement_orders[mode_sla_failure_rate])
```

2.) Delivery Delay Probability

```
1 Delay Probability =  
2 AVERAGE(fact_procurement_orders[delay_flag])
```

3.) Lead Time Stability Score

```
1 Lead Time Stability Score =  
2 AVERAGE(fact_procurement_orders[lead_time_variance])
```

4.) SLA Failure Rate

```
1 SLA Failure Rate =  
2 AVERAGE(fact_procurement_orders[sla_failure_flag])
```


KPI ENGINEERING (Power BI – DAX)

5.) Dynamic Risk Status

```
1 Dynamic Risk Status =
2 VAR AvgRisk =
3     CALCULATE(
4         AVERAGE(fact_procurement_orders[delay_flag]),
5         ALL(fact_procurement_orders)
6     )
7
8 VAR SelectedRisk =
9     [Supplier Risk Index]
10
11 RETURN
12 SWITCH(
13     TRUE(),
14     SelectedRisk <= AvgRisk * 0.9, "Low Risk",
15     SelectedRisk <= AvgRisk * 1.1, "Moderate Risk",
16     "High Risk"
17 )
```

6.) Risk Adjusted Cost

```
1 Risk Adjusted Cost =
2 AVERAGE(fact_procurement_orders[cost_per_unit])
3 *
4 (1 + [Delay Probability])
```

7.) Region-Wise Procurement Cost

```
1 Region-wise Procurement Cost =
2 CALCULATE(
3     AVERAGE(fact_procurement_orders[cost_per_unit]),
4     ALLEXCEPT(
5         fact_procurement_orders,
6         fact_procurement_orders[order_region]
7     )
8 )
```


DATA MODELING

Modeling Approach

To enable reliable procurement risk analysis, a dimensional data model was implemented in Power BI using a Star Schema Architecture.

Fact Table

fact_procurement_orders containing:

- Order Date
- Shipment Mode
- Procurement Region
- Lead Time
- Delay Flag
- SLA Failure Flag
- Cost per Unit

DATA MODELING

Dimension Tables Created

- dim_region
- dim_shipping_mode
- dim_category

Relationships Established:

dim_region → fact_procurement_orders

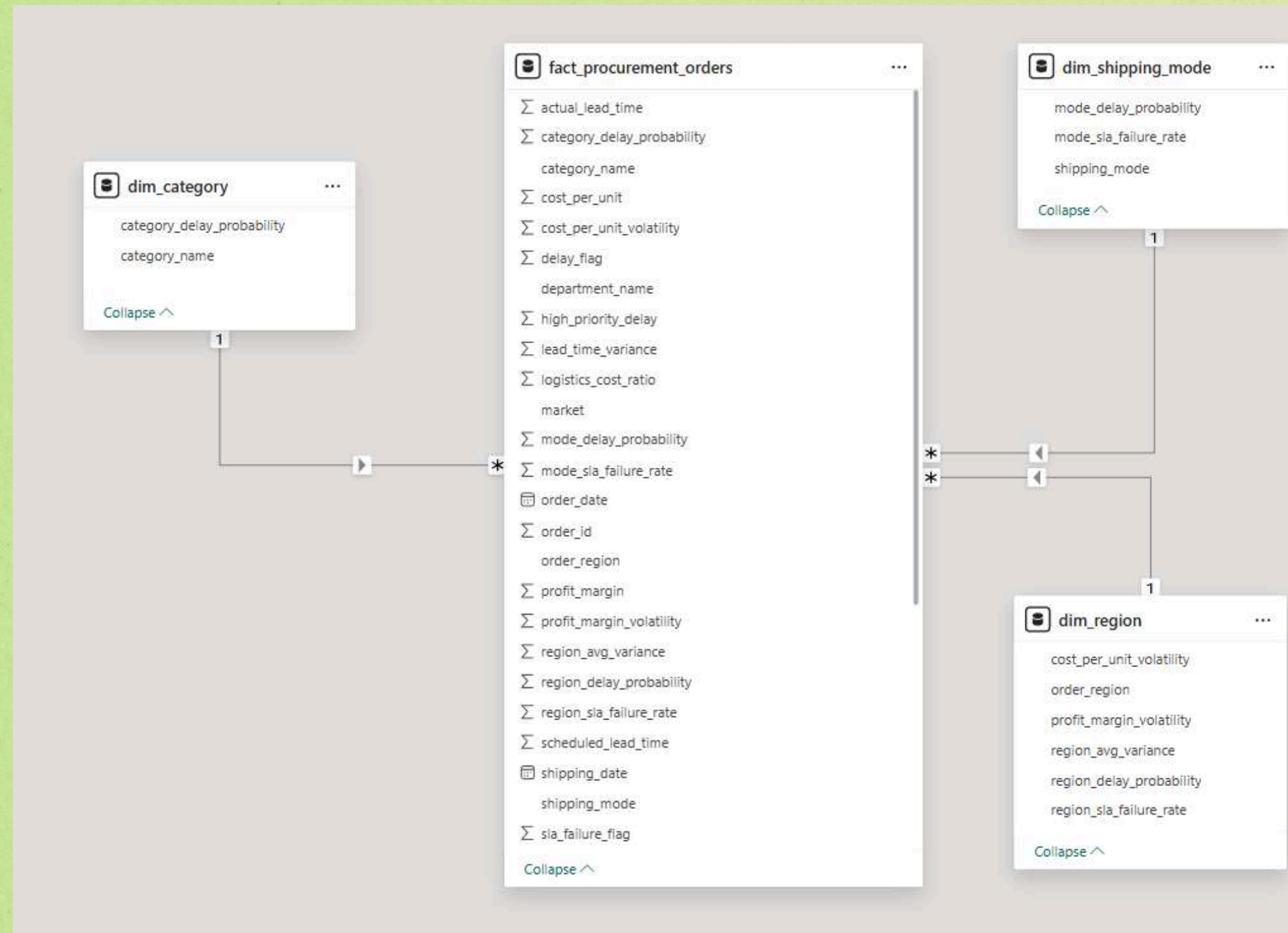
dim_shipping_mode → fact_procurement_orders

dim_category → fact_procurement_orders

Modelling Outcome:

Implementation of the star schema enabled:

- Efficient region-wise risk analysis
- Shipment mode performance benchmarking
- Reliable category-level KPI computation
- Improved dashboard responsiveness



DASHBOARD OVERVIEW

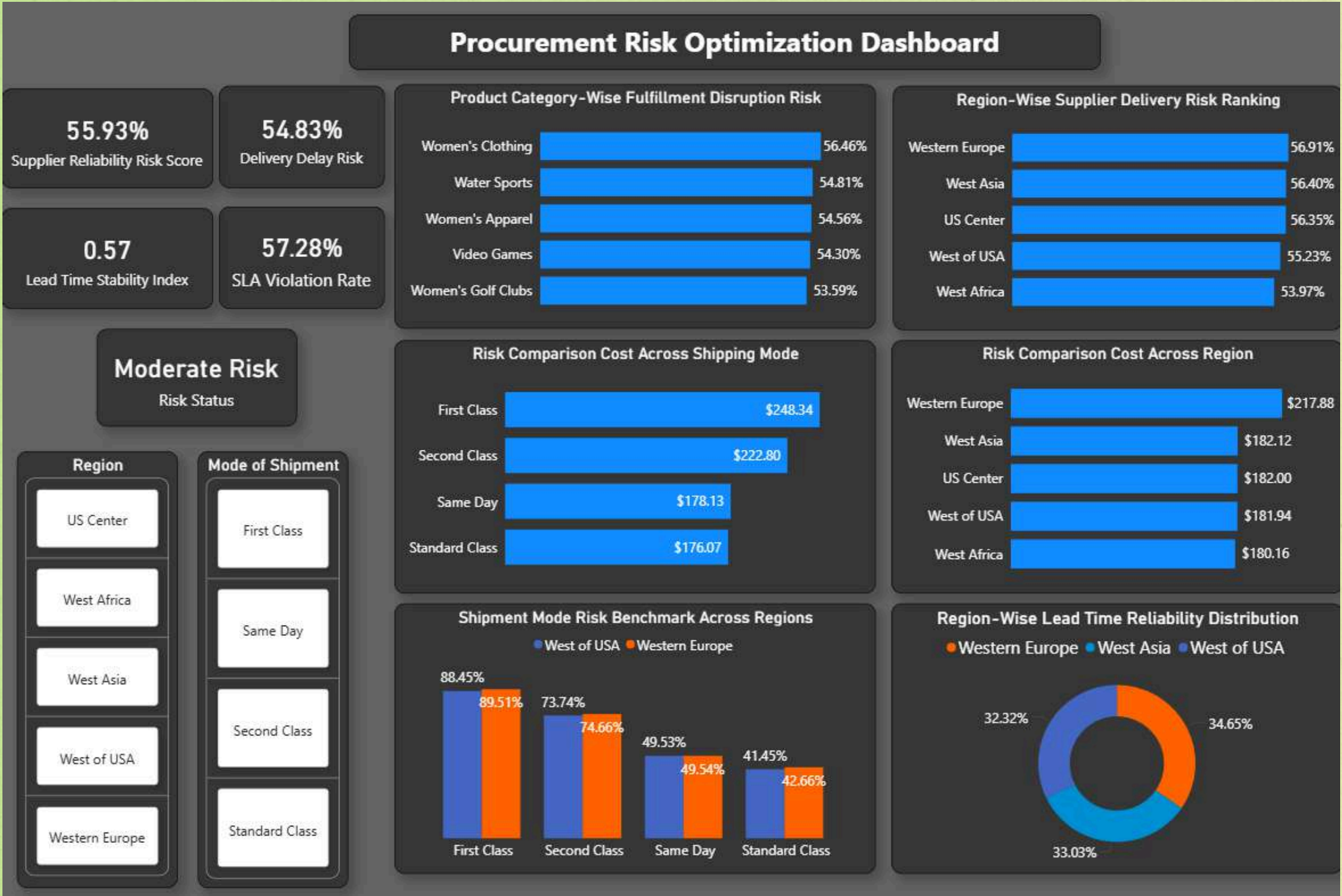
Procurement Risk Optimization Dashboard

Objective:

To evaluate supplier reliability across sourcing regions and identify shipment strategies impacting procurement cost and SLA compliance.

Analysis Focus:

- Category-wise Delivery Delay Risk
- Region-wise Supplier Risk Ranking
- Shipment Mode Risk Benchmark
- Region-wise Procurement Cost
- Lead Time Stability Distribution



DASHBOARD INSIGHTS

Insight 1:

Region-wise supplier risk analysis indicates that procurement transactions sourced from specific regions exhibit consistently higher delivery delay probability and SLA violation rates.

This suggests the presence of supplier-level fulfillment inefficiencies and operational bottlenecks within these sourcing networks, potentially impacting procurement timelines and downstream inventory availability.

Recommendation 1:

Procurement operations should prioritize supplier reliability benchmarking across sourcing regions and initiate targeted supplier performance audits in risk-prone regions to ensure compliance with delivery schedules and contractual SLA commitments.

DASHBOARD INSIGHTS

Insight 2:

Shipment mode benchmarking reveals that premium logistics options such as First Class incur significantly higher procurement cost per unit without proportionate improvement in delivery reliability or SLA compliance across multiple sourcing regions.

Recommendation 2:

Expedited shipment modes should be selectively deployed for procurement transactions with elevated delay risk or time-sensitive operational requirements, while cost-efficient shipment modes may be adopted for stable sourcing categories to optimize procurement expenditure.

DASHBOARD INSIGHTS

Insight 3:

Comparative regional analysis indicates that sourcing regions exhibiting higher procurement cost do not necessarily demonstrate improved supplier reliability or reduced delivery risk, highlighting the absence of a direct correlation between procurement spend and sourcing performance.

Recommendation 3:

Procurement strategies should integrate region-wise supplier reliability assessment into vendor selection frameworks to ensure sourcing decisions are aligned with both cost efficiency and operational fulfillment performance.

DASHBOARD INSIGHTS

Insight 4:

Lead time stability distribution across procurement regions highlights variability in fulfillment consistency, indicating region-dependent uncertainty in supplier delivery timelines that may impact procurement planning and inventory replenishment cycles.

Recommendation 4:

Institutionalizing periodic monitoring of supplier lead time performance can enable proactive identification of sourcing disruptions and facilitate region-specific procurement strategy adjustments to improve delivery consistency.

DASHBOARD INSIGHTS

Insight 5:

Category-wise fulfillment disruption analysis suggests that certain product categories are more susceptible to delivery delays under varying shipment modes and sourcing regions, indicating the need for category-specific procurement strategies.

Recommendation 5:

Procurement planning should incorporate product category sensitivity to delivery risk by aligning shipment mode selection and supplier sourcing strategy with category-level fulfillment requirements to ensure cost-effective and reliable procurement operations.

BUSINESS IMPACT

Operational Impact:

Implementation of supplier reliability-driven procurement strategies can enable improved sourcing consistency across procurement regions by minimizing delivery delays and SLA violations associated with high-risk suppliers.

Financial Impact:

Adopting risk-aware shipment mode selection can reduce procurement cost per unit by avoiding over-reliance on premium logistics options that do not yield proportional improvements in fulfillment performance.

Strategic Impact:

Incorporating region-wise supplier risk benchmarking into vendor selection frameworks allows procurement teams to align sourcing decisions with operational reliability, thereby mitigating fulfillment uncertainty across procurement networks.

Planning Impact:

Monitoring lead time stability across sourcing regions enables procurement planners to proactively identify fulfillment variability and adjust sourcing strategy to ensure procurement schedule adherence.

Category-Level Impact:

Category-specific disruption risk assessment supports the development of differentiated procurement strategies tailored to product-level delivery sensitivity, improving sourcing effectiveness across high-risk categories.

BUSINESS IMPACT

Implementation of data-driven procurement optimization practices may enable the organization to:

- Improve supplier delivery reliability across sourcing regions
- Reduce procurement delays impacting operational timelines
- Minimize SLA violations through shipment mode optimization
- Optimize procurement expenditure by aligning sourcing strategy with supplier performance
- Improve lead time consistency in procurement operations
- Support cost-efficient vendor selection decisions

CONCLUSION

This project evaluated supplier delivery performance and procurement cost dynamics across sourcing regions to identify shipment strategies and sourcing networks contributing to operational risk exposure.

By integrating supplier reliability assessment into procurement planning processes, the organization can adopt risk-constrained sourcing strategies that balance procurement cost efficiency with fulfillment consistency.

Adoption of data-driven procurement decision frameworks can enable the organization to:

- Enhance sourcing reliability
- Improve SLA compliance
- Reduce logistics expenditure
- Maintain procurement schedule consistency
- Support strategic supplier selection